Cotton Yield Prediction & Price Trend Analysis (Maharashtra) - Summary

From data collection to a simple model | Author: Jayesh Patil

1. Question

Can monsoon rain help estimate Maharashtra's cotton yield (kg per hectare), and what do cotton prices tell us over time?

2a. Prices

- Source: Agmarknet reports
- 30 dates (Oct-Dec), 2021-2026
- Maharashtra weighted average

2b. Yield

- Source: RBI Handbook, Table 86
- 21 seasons (2004-05 to 2024-25)

2c. Weather

- Source: NASA POWER (monthly)
- Jalgaon, Akola, Yavatmal
- Rain and temperature, 2004-2025

3. Cleaning and joining (Python / pandas)

combine_prices.py → make_yield.py → build_season_data.py → **season_data.csv** (21 seasons, no missing rows)
Per season: yield + June-September rain + temperature. World cotton price (World Bank) added for the price comparison.

4. Observations (charts and correlation)

Rain ↔ yield	**+0.38**	mild positive link
Temperature ↔ yield	**-0.26**	weak negative link
Rain ↔ temperature	**-0.50**	drier years are hotter
World price ↔ Indian price	**+0.57**	on 20 well-supplied dates

5. Model and honest testing

Linear Regression + **Leave-One-Out** (hide one season, predict it, repeat 21 times)

Prediction	Average error (kg/ha)
Always guess the average (baseline)	52.7
Monsoon rain only	**49.0**
Rain + temperature	51.3

Examples (about +/- 49 kg/ha): dry year ~263, normal year ~302, wet year ~339 kg/ha.

6. Conclusions and limitations

- Rain-only model is slightly better (about 7%) than guessing the average, but it cannot give precise forecasts.
- Only 21 seasons; pests, seed type, irrigation and area are not in the data. Adding temperature did not help.
- No price model: only 30 price dates. World price did not predict Indian price when a whole year was held out.
- Next: long monthly Indian price history, district-wise yield, more factors.

Sources: Agmarknet, RBI Handbook of Statistics on Indian States (Table 86), NASA POWER, World Bank Commodity Markets (Pink Sheet).